

Polar Plots

Tracing $G(j\omega)$ in the complex plane.

Ogata, *Modern Control Engineering*, Ch. 7.

In this tutorial you will:

- plot the polar locus of a first-order and a type-1 system by hand,
- verify it against `nyquist`, and
- see how loop gain scales the locus toward the critical point -1 .

Step through with **Ctrl+Enter**, or render a report with **publish**.

The polar plot of $G(j\omega)$ traces the locus of the complex number $G(j\omega) = \text{Re } G(j\omega) + j \text{Im } G(j\omega)$ in the complex plane as ω sweeps from 0 to ∞ . Unlike the Bode diagram, magnitude and phase are shown on a single plot, at the cost of frequency no longer being an explicit axis (it is only a parameter along the curve).

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Worked Example: First-Order System

$$G(s) = \frac{1}{1+j\omega T}, T = 1.$$

$$G(j\omega) = \frac{1}{1+\omega^2 T^2} - j \frac{\omega T}{1+\omega^2 T^2}$$

At $\omega = 0$: $G = 1 + j0$. As $\omega \rightarrow \infty$: $G \rightarrow 0 - j0$. This traces a semicircle in the lower half-plane, centered at $(0.5, 0)$ with radius 0.5 – a classical result, verified below algebraically.

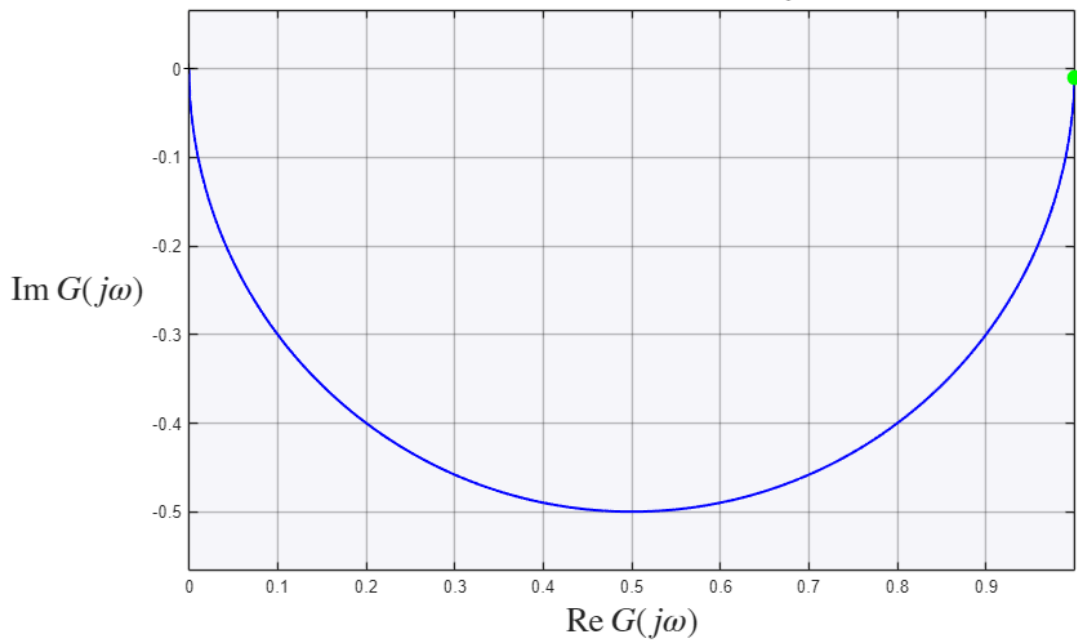
```
T = 1;
w = logspace(-2,3,1000);
Gjw = 1./(1+1j*w*T);
re = real(Gjw); im = imag(Gjw);

% Verify it lies on the circle (re-0.5)^2 + im^2 = 0.25
resid = (re-0.5).^2 + im.^2 - 0.25;
fprintf('Max deviation from ideal semicircle: %.2e\n', max(abs(resid)))

figure
plot(re,im,'b','LineWidth',1.5)
hold on
plot(re(1),im(1),'go','MarkerSize',8,'MarkerFaceColor','g')
plot(0,0,'k+')
hold off
axis equal
grid on
title('Polar Plot:  $G(j\omega) = \frac{1}{1+j\omega}$ ','Interpreter','latex','FontSize',20)
ylabel('j\mathrm{Im}\{G(j\omega)\}','Interpreter','latex','FontSize',20)
set(get(gca,'YLabel'),'Rotation',0)
xlabel('j\mathrm{Re}\{G(j\omega)\}','Interpreter','latex','FontSize',20)
```

```
Max deviation from ideal semicircle: 1.67e-16
```

$$\text{Polar Plot: } G(j\omega) = \frac{1}{1 + j\omega}$$



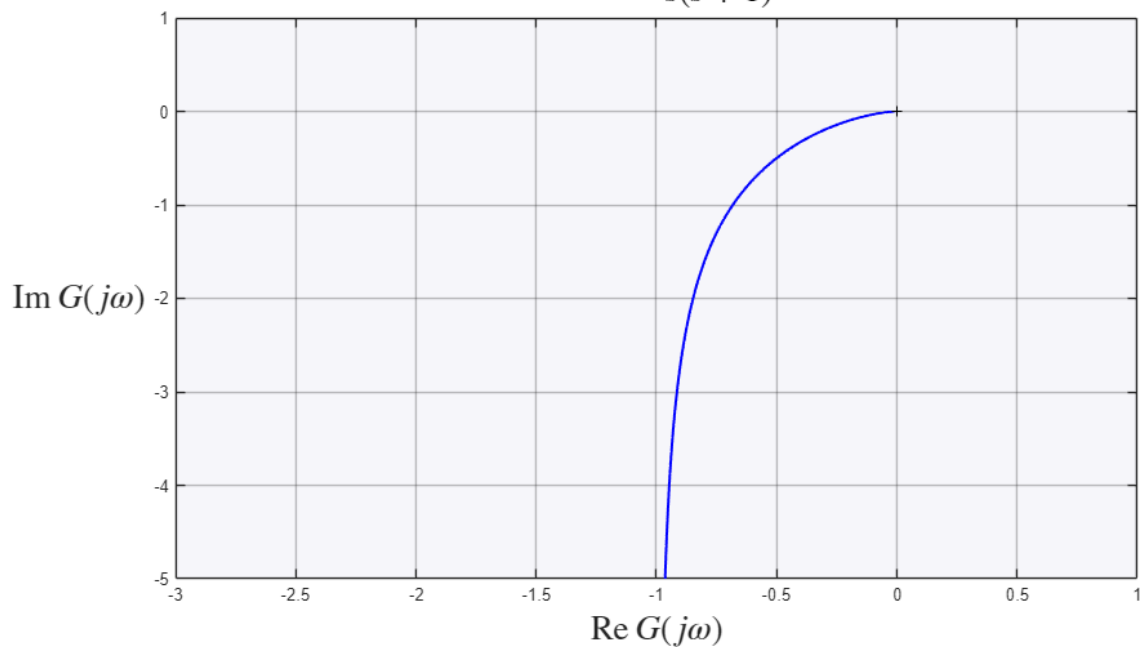
Worked Example: Type-1 System

$G(s) = \frac{K}{s(Ts+1)}$, $K = 1, T = 1$. As $\omega \rightarrow 0$, $G(j\omega) \rightarrow \infty \angle -90^\circ$ (the locus runs off to $-j\infty$ along the vertical asymptote $\text{Re} = -KT = -1$); as $\omega \rightarrow \infty$, $G(j\omega) \rightarrow 0 \angle -180^\circ$.

```
G2 = tf(1,[1 1 0]);
w2 = logspace(-2,2,500);
Gjw2 = squeeze(freqresp(G2,w2));

figure
plot(real(Gjw2),imag(Gjw2),'b','LineWidth',1.5)
hold on
plot(0,0,'k+')
hold off
grid on
xlim([-3 1]); ylim([-5 1])
title('Polar Plot:  $G(s)=\frac{1}{s(s+1)}$  (Type-1)', 'Interpreter','latex', 'FontSize',20)
ylabel('Im  $G(j\omega)$ ', 'Interpreter','latex', 'FontSize',20)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('Re  $G(j\omega)$ ', 'Interpreter','latex', 'FontSize',20)
```

Polar Plot: $G(s) = \frac{1}{s(s+1)}$ (Type-1)

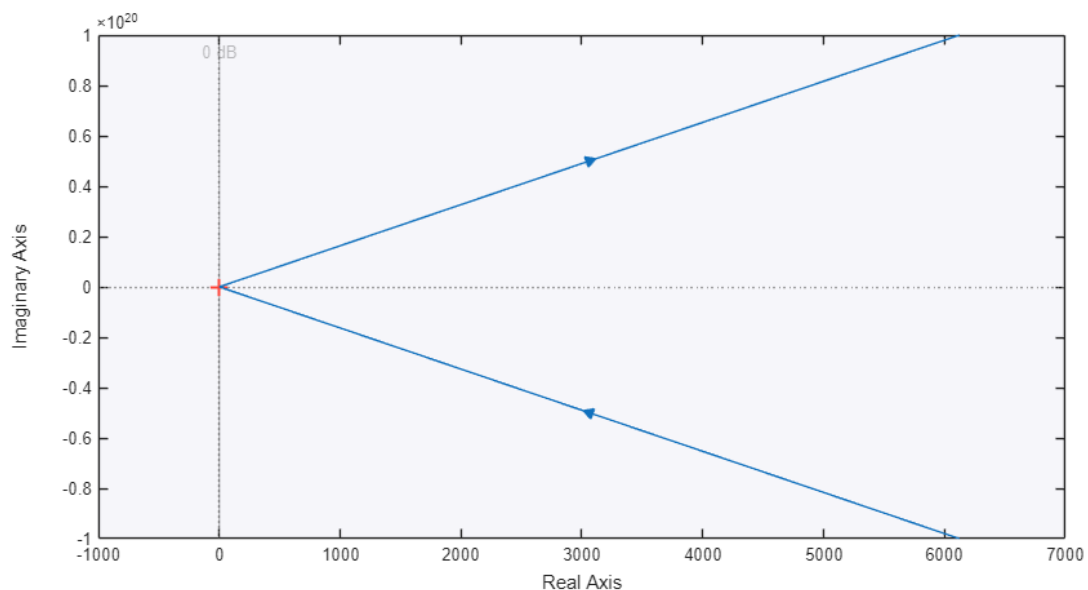


Using the Control System Toolbox Directly

`nyquist` automates the polar-plot construction (including negative frequencies, shown as the mirror image by symmetry).

```
figure
nyquist(G2)
title('Nyquist Plot via |nyquist|:  $G(s)=\frac{1}{s(s+1)}$ ','Interpreter','latex','FontSize',20)
grid on
```

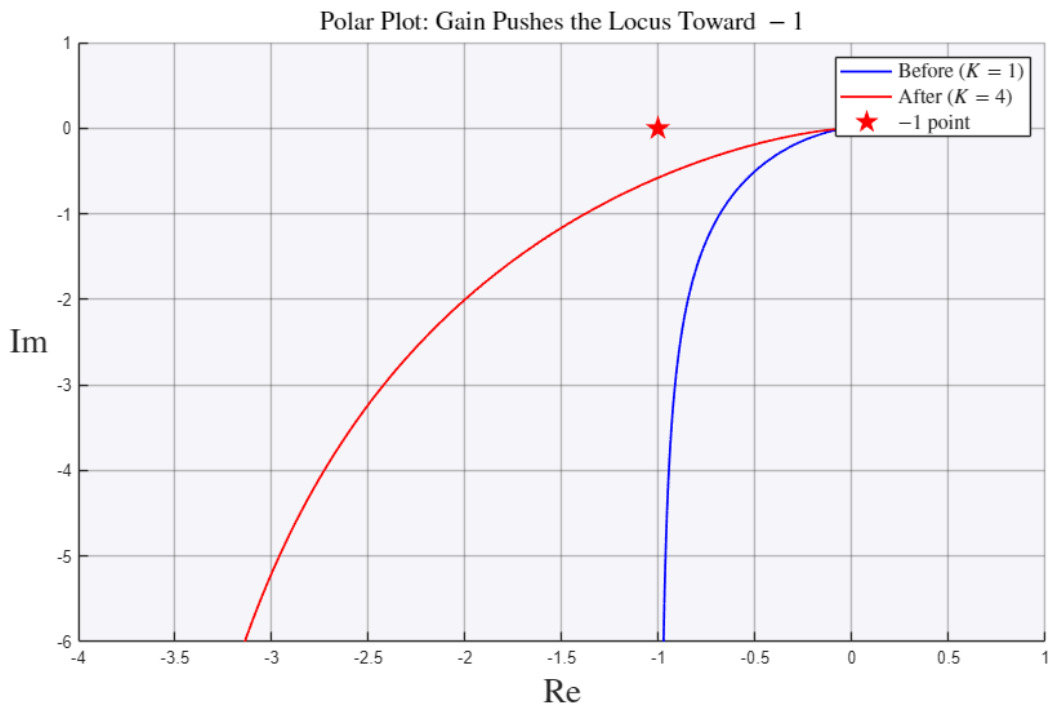
Nyquist Plot via `nyquist`: $G(s) = \frac{1}{s(s+1)}$



Before vs. After: Gain Scales the Polar Plot Toward -1

Raising the loop gain scales the whole polar plot radially while the -1 point (red) stays fixed, so a higher gain pushes the locus closer to (and eventually around) -1 – the geometric onset of instability that the Nyquist criterion formalizes.

```
G2a = tf(1,[1 1 0]); G2b = tf(4,[1 1 0]);
figure
hold on
plot(real(squeeze(freqresp(G2a,w2))),imag(squeeze(freqresp(G2a,w2))), 'b', 'LineWidth', 1.3)
plot(real(squeeze(freqresp(G2b,w2))),imag(squeeze(freqresp(G2b,w2))), 'r', 'LineWidth', 1.3)
plot(-1,0, 'rp', 'MarkerSize', 13, 'MarkerFaceColor', 'r')
hold off
grid on
xlim([-4 1]); ylim([-6 1])
legend('Before ( $K=1$ )', 'After ( $K=4$ )', '$-1$ point', 'Interpreter', 'latex', 'FontSize', 12)
title('Polar Plot: Gain Pushes the Locus Toward  $-1$ ', 'Interpreter', 'latex', 'FontSize', 15)
ylabel('$\mathrm{Im}$', 'Interpreter', 'latex', 'FontSize', 20)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$\mathrm{Re}$', 'Interpreter', 'latex', 'FontSize', 20)
```



Try it yourself

- Raise the type-1 gain from 4 toward 8 and notice the locus swing closer to the -1 point – the geometric approach to instability.
- Increase T in the first example and watch the semicircle radius stay 0.5 but the frequencies redistribute along it.